

and even if subject to change, they reveal quite an interesting picture of what forms the face of the Larrabee architecture.

The first thing to understand is that 'Larrabee' is codename for an umbrella of products. Much like Core 2 Duo and Core 2 Extreme fall under the same multi-core banner of Intel marketing; Larrabee will consist of a series of implementations, sharing a common architecture. Differences in the detail will make one Larrabee product suitable for notebooks, another for a desktop graphic card, and yet another suitable for use in a transaction server.

The 'many-core' detail of this new architecture will be realized by 16-24 cores. Each core will run between 1.7 and 2.5 GHz depending upon the product that the processor will form. The minimum heat signature of the implementation will run at least 150W. The number of cores on each chip, as well as the clock speed, will depend on which product the architecture will form the basis of: integrated into the motherboard, high-end GPU, transactions co-processor, and so on.

Each of the cores will be capable of processing four simultaneous threads of execution. Each core will also come with an L1 cache as well as a floating point unit (FPU) which will implement a subset of the x86 instruction set. The x86 set is an important point of differentiation between the Larrabee implementation and current GPU implementations—a graphics programmer uses special instruction code to do what needs to be done via a GPU. This instruction set is hidden behind the more user-friendly API of either OpenGL or DirectX layers. While Larrabee will retain OGL and DX compatibility, it will also introduce a specific instruction set called the Advance Vector Extension (AVX) which will help leverage the FPU potential of the architecture.

The AVX set is to be seen in a similar light to SSE in current x86 processors—a "256-bit vector extension to SSE, for floating-point intensive applications".

This software soup of x86 instructions and extensions is what Intel is betting on.

They hope that a more familiar set of code will encourage programmers to make better use of the floating point capabilities of the Larrabee architecture. This line of thinking could be construed as a bit naïve, considering the fact that very few games have made use of the HyperThreading capabilities of Intel processors; and that too, only recently.

Back to the Larrabee architecture—the many-core processor will consist of a large L2 cache. This pool of L2 will be shared by each core of the processor. Furthermore, each core will also be able to lock a part of the L2 cache for its own use. Larrabee will also see an on-die memory controller (the memory controller is moving on-die for a variety of Intel offerings, similar to what AMD already has). The final piece of the Larrabee die will determine the nature of the final product. This final piece is a fixed-function unit, dedicated to a single task. A Larrabee GPU will have a texture sampler for this fixed-function unit (a texture sample is essentially one-stage of the DX / OGL pipeline), a Larrabee server coprocessor will have a hardware encryption unit there, and so on.

All these components will be connected together via a ring bus, similar to what is used by IBM's Cell.

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